Abdelrahman Mohamed

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Calgary, AB | Canadian citizen, TN-eligible for US roles (no H-1B required) | Open to relocation

Education

University of Calgary — Calgary, AB
Bachelor of Science in Computer Science — *Expected Dec 2026*
Relevant Coursework: Web-Based Systems, Database Management Systems, Data Structures & Algorithms, Design & Analysis of Algorithms, Software Engineering, Operating Systems, Computer Networks
Honours & Interests: Dean's List, Faculty of Science (Winter 2023); Chess.com peak rating 2000+ (blitz, bullet)

Experience

CYD Soccer Academy — May 2025 – Present
Software Engineer & Technical Partner | cydsoccer.com — *Calgary, AB*
- Sole engineer on the academy's operating system: a Next.js 15/TypeScript application on Vercel spanning ∼90 API routes and 426 source files, used by roughly 500 families across four Calgary venues.
- Built the registration wizard and Stripe Checkout integration covering one-time, instalment, and subscription payments, with a scheduled job enforcing instalment caps and a referral system that applies a discount at checkout and refunds the referrer, made idempotent per referrer-friend pair.
- Diagnosed a production Twilio fault where smart apostrophes and emoji silently forced messages out of GSM-7 into UCS-2, halving per-segment capacity; added input sanitization and pre-send segment estimation, on a channel where carrier segment fees are roughly 47% of messaging cost.

City of Calgary — Sep 2025 – Apr 2026
Software Engineer, Capstone Project — *Calgary, AB*
- Built an automated ETL pipeline that centralized corporate greenhouse-gas activity data from 4 business units into SQL Server, replacing a manual spreadsheet consolidation process.
- Implemented Python ingestion and transformation jobs on Azure with validation steps that surface source-data inconsistencies, delivering results through Power BI dashboards used by municipal staff for emissions reporting.

Mercor — Nov 2025 – Jan 2026
Software Engineering Expert, AI Training — *Remote — San Francisco, CA*
- Selected through technical screening into Mercor's expert software engineering track; authored agentic coding evaluations from public GitHub repositories spanning CLI tooling, data science, and full-stack applications, writing intentionally underspecified change requests and scoring model attempts on correctness and performance.

Customer Maps — Mar 2025 – Jun 2025
AI/ML Intern — *Calgary, AB*
- Built retail-facing AI agents on Google Vertex AI and the Agent Development Kit as part of the Google Innovate Program cohort, automating a recurring stakeholder reporting workflow.
- Designed GCP data pipelines over BigQuery spanning 100K+ records and built React dashboards for stakeholder review.

DATech — Apr 2024 – Mar 2025
AI Evaluation Contractor (RLHF) — *Remote — New York, NY*
- Reviewed and scored 1,000+ model-generated code samples across Python, JavaScript, and SQL against correctness, security, and style rubrics, producing labeled preference data for RLHF training pipelines and writing reference solutions to probe code-generation failure modes.

Projects

Systemlab | *TypeScript, Next.js, React Flow, Web Workers, Gemini API, Supabase, Vitest*
- Built a deterministic discrete-event simulation engine (seeded RNG, lognormal service times) modeling replication lag, quorums, sharding, keyed caches with coalescing, retries, circuit breakers, and region outages, run in a Web Worker for reproducible, non-blocking runs; simulates 180K requests in under 2 seconds.
- Authored a 70-lesson system design curriculum whose reference solutions are test-verified across seeds (946 Vitest tests), plus a Gemini-graded design defense and voice mock interview scored against rubrics.

Road Object Detection *(CPSC 599, team of 3)* | *Python, Ultralytics YOLOv8/v11, PyTorch*
- Fine-tuned and benchmarked YOLOv8s against YOLOv11m on a 12-class Egyptian traffic dataset containing categories absent from Western autonomous-driving benchmarks (tuk-tuks, riders, street banners); owned the v11 training arm and the accuracy-vs-cost comparison across parameters, FLOPs, and latency behind the final model recommendation.

Technical Skills

Languages: Python, TypeScript/JavaScript, SQL, Java
Frameworks & Data: Next.js, React, Node.js, Supabase, PostgreSQL, SQL Server, Prisma, Stripe, REST APIs
Cloud & Infrastructure: GCP (Vertex AI, BigQuery), Azure, Vercel, Docker, Git, CI/CD, Power BI
AI/ML: LLM & RAG application design, RLHF evaluation, prompt engineering, computer vision (YOLOv8/v11, OpenCV, PyTorch); Claude Code, Cursor, MCP